

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #56 – Peer Skill Exchange & Mentorship Network

Scenario: A time-banking skill exchange platform where community members trade skills. For example, one hour of Python tutoring may be exchanged for one hour of Guitar lessons. The platform maintains time credits and supports mutual feedback and rating validation.

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall maintain a time-credit balance for each member and credit 1 hour to a mentor upon verified completion of a teaching session.	High	Pass: After both participants sign off a completed 1-hour session, the mentor's balance increases by 1 credit and the learner's balance decreases by 1 credit. Fail: An unverified session does not transfer credits.	Time credits are the core mechanism that enables fair skill exchange.
FR-002	Functional	The system shall allow a member to create and manage a skill profile by adding skills they can teach and skills they want to learn.	High	Pass: A member can add, edit, or remove at least one offered skill and one requested skill, and the updated profile is displayed after saving. Fail: Invalid skill entries are not saved.	Skill profiles allow members to discover suitable learning and mentoring opportunities.
FR-003	Functional	The system shall allow a learner member to search available skill mentors and request a teaching session for a selected skill and time slot.	High	Pass: A learner can view matching mentors, select a mentor and an available time slot, and submit a session request. Fail: The system prevents a request for an unavailable time slot.	Session discovery and booking are essential for members to exchange skills.
FR-004	Functional	The system shall allow a learner and skill mentor to confirm the completion of a scheduled teaching session.	High	Pass: When both participants confirm completion, the session status changes to 'Verified Completed' and triggers the time-credit update defined in FR-001. Fail: If either participant has not confirmed, no credit transfer occurs.	Mutual confirmation provides validation before time credits are transferred.
FR-005	Functional	The system shall allow members to submit a rating and feedback for the other participant after a verified completed session.	Medium	Pass: After a verified session, each participant can submit one rating and feedback entry for the other participant. Fail: Feedback cannot be submitted for an unverified session.	Feedback and ratings help build trust and support quality mentoring within the community.
NFR-001	Nonfunctional – Performance & Security	The system shall provide session scheduling calendar synchronization with Google Calendar and Outlook through iCal export feeds while meeting defined performance and security standards.	High	Pass: Benchmark tests confirm target latency and security standards under simulated peak load, and generated iCal feeds can be imported by supported calendar applications. Fail: The system fails defined tests or produces an invalid calendar feed.	Reliable calendar integration and secure operation support dependable session scheduling.
NFR-002	Nonfunctional – Usability	The system shall provide clear feedback messages for successful bookings, verification status, credit transfers, and validation errors.	Medium	Pass: Each tested action displays a clear success, status, or error message, and users can identify the next required action. Fail: A tested action provides no understandable feedback.	Clear system feedback reduces confusion during scheduling, verification, and time-credit exchanges.

2. Use-Case Flow Specification

Use Case: UC-03 – Book and Complete Skill Exchange Session

Primary Actor: Learner Member

Supporting Actor: Skill Mentor

Preconditions

- The learner member and skill mentor have active member accounts.
- The mentor has listed the relevant skill and an available time slot.
- The learner has at least 1 time credit available for the requested 1-hour session.

Postconditions

- If the session is verified as completed, the session status is updated to 'Verified Completed'.
- The time-credit transfer is recorded: the learner is debited and the mentor is credited according to the verified session duration.
- Both participants become eligible to submit a rating and feedback for the completed session.

Main Success Scenario

1. The learner searches for mentors offering the desired skill.
2. The system displays matching skill mentors and their available time slots.
3. The learner selects a mentor and an available 1-hour time slot.
4. The learner submits a session request.
5. The system checks the learner's available time-credit balance and the selected slot availability.
6. The system confirms the booking and records the scheduled session.
7. The learner and mentor attend the scheduled skill exchange session.
8. After the session, the learner confirms that the session was completed.
9. The mentor confirms that the session was completed.
10. The system verifies that both participants have confirmed completion.
11. The system updates the time-credit balances and marks the session as 'Verified Completed'.
12. The system notifies both participants that the session was successfully verified.
13. The use case ends successfully.

Alternate Flow – 5a. Insufficient Time Credits

- 5a1. The system detects that the learner does not have sufficient time credits for the requested session.
- 5a2. The system displays an 'Insufficient time credits' message and does not confirm the booking.
- 5a3. The learner may cancel the request and earn additional credits by completing a teaching session.
- 5a4. The use case ends without creating the requested booking.

Peer Skill Exchange & Mentorship Network – Use-Case Diagram

